3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

The NEPTUNE Project: An Interactive Earth-Ocean Observatory at the Scale of a Tectonic Plate

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ABSTRACT

The NEPTUNE project will deploy a submarine cabled network on the Juan de Fuca tectonic plate off the U.S. and Canadian west coasts. Using fiber-optic/power cable and junction boxes, this plate-scale ocean observatory will establish a linked array of undersea observatories by providing significant amounts of power and an Internet communications link to sensors and sensor networks on, above, and below the sea floor. NEPTUNE's goal is to enable real-time, long-term, plate-scale studies in the ocean and earth sciences. The network may also serve as a unique test bed for sensor and robotic systems designed to explore other oceans in the solar system. NEPTUNE will focus on a broad range of scientific inquiries ranging from research involving microbial productivity from the oceanic crust related to deformation and volcanic eruptions in a plate tectonic framework, to abroad array of earthquake and tsunami-related issues, to tracking fish migration, to observing sediment transport, to monitoring carbon fluxes related to air-sea exchange and primary productivity in the upper ocean, including earthquake generated outgassing of subduction complexes. Hard-wired to the Internet, NEPTUNE will enable a new generation of ocean and earth science to be conducted for decades from instrument arrays immersed in the dynamic environments within and beneath our oceans. It will also bring real-time data and imagery into the laboratories, classrooms and living rooms connected to the Internet. For the first time, the public will be able to electronically watch over the shoulders of oceanographers and geoscientists as they explore the time domain within the ocean basins of our planet.